차준하 (Junha Cha)

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Department of AI
Gachon University

Education

University	**Gachon University,** Department of AI & Software Expected graduation: February 2027
GPA	4.35/4.5 | Major : 4.37/4.5
Military	육군 병장 만기전역, ROKA (Jan 2022 - Jul 2023)

Activity

Research Intern **26.01~present**	**Korea University** **Robot Intelligence Lab (RILAB)** - Research focused on Tele-operation, Force Estimation, and Dexterous Manipulation for achieving Human-Level Intelligence in robotics.
Intern **25.06~25.12**	**ROBOTIS AI Software Team Intern** - Contributed to the design and coding of AI Worker and OMX modules as a member of the AI Software Team.
Research Assistant **23.12~25.06**	**Gachon University** **Intelligent Robotics and Autonomous System Control Lab (iRASC)** - Research focused on Intelligent Manipulation for achieving Human-Level Intelligence in robotics.

Domestic Conference

2025 **대한전자공학회,IEIE(Robot Intelligence) :**

- Junha Cha, et al., "BB-VLA: A Brain-Body Framework Mimicking Human Thought for Intelligent Manipulator Palletizing."

Project

2025 **VR Teleoperation of Humanoid Robots with PAPRLE**

- Developed a demo set for the AI Worker Humanoid by tuning and implementing UIUC KIMLAB's open-source PAPRLE framework.

2025 **Main Contributor for ROBOTIS 'OMX' Product Launch**

- Led and contributed to the entire product launch cycle of 'OMX' (a ROBOTIS Physical AI product), including market research, hardware, software, CS, manual preparation, demo set creation, and release management.

2024 **Team Project: Low Scale Smart-Factory System**

- Developed a smart-factory system, where a robotic arm transports items based on ArUco Marker recognition, ordered by customers via a web-based interface.

2024 **국립재활원 과제 : Feedback System for Personalized Rehabilitation Robots**

- Reconstructed the existing patient experiment data acquisition system using ROS, and developed a ROS driver for the IMU sensor.

2025 **Team Project: Remotix**

- Developed 'Remotix,' a web-based customer support system for autonomous driving, integrating a remote teleoperation interface using a Turtlebot3 platform.

2021 **Team Project: Autonomous Car and Smart City**

- Implemented an autonomous vehicle that detects proximity and navigates using an Arduino ultrasonic sensor kit, and designed the associated smart city infrastructure.

Awards

2025 **Extreme Robotics Challenge, Organized by: Korea Institute of Robotics and Convergence(KIRO) – Second Place**

- Role : System Integration(mission controller) and Manipulation
- Topic : Button pressing, rescue supply collection, and door opening missions in a simulated harsh disaster environment

2025 **Defense Robotics Competition, Organized by: Korea Institute of Robotics and Convergence(KIRO) – Honorary Mention**

- Role : Leader-Follower System Construction and Manipulation
- Topic : Collected supply boxes discovered during UGV navigation with a robotic arm, demonstrating leader-follower coordination.

2024 **DroneBot Challenge, Organized by: Ministry of National Defense(ROK) - First Place**

- Role : PyMavlink value WebSocket Communication, System Dummy Data Development, and Data Curation
- Topic : Proposed an AI-Vision-based runway Foreign Object Debris(FOD) detection platform for emergency combat aircraft deployment

2024 **Defense Robotics Competition, Organized by: Korea Institute of Robotics and Convergence(KIRO) – Robotics Industry Award**

- Role : Developed enemy recognition system and track segmentation using Jetson Orin and YOLOv8
- Topic : Developed a multi-functional autonomous driving system with an attached robotic arm for military operations

2021 **창업해 春, Organized by: Gachon University – First Place**

- Role : Market Research, MD, and Frontend Development using Flutter
- Topic : Developed the university student community application, **'Univerlet'** during the 6-months challenge

Research Interest

- Tele-operation
- Force Estimation
- Dexterous Manipulation

Skill

- ROS
- Programming Skill (C, Python)
- Docker
- Git
- Robotics